Product Information

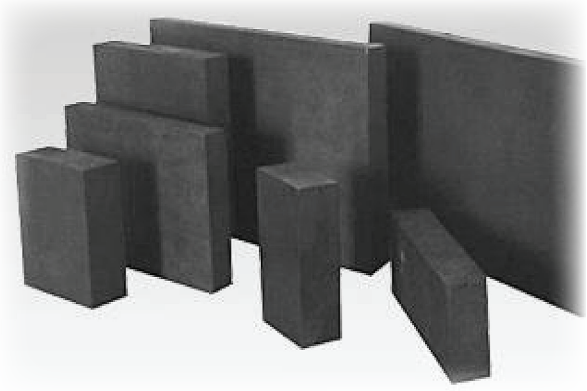
Graphite Block

Dry Film Lubricant Guarantees High Performance in Hostile Environments while Reducing Wear and Noise

Graphite Block dry film lubricant is composed of synthetic, lubricant-quality graphite that is specifically designed to lubricate trunnion rollers, riding rings and insert seals on rotary kilns, calciners, dryers and other industrial equipment requiring dry film lubrication.

The Graphite Block lubricant lowers pressure on the bearing thrust mechanism, while reducing forces on bearing end caps or thrust collars. It substantially reduces wear and noise, resulting in decreased maintenance costs; smoother, quieter operation; and longer lasting equipment.

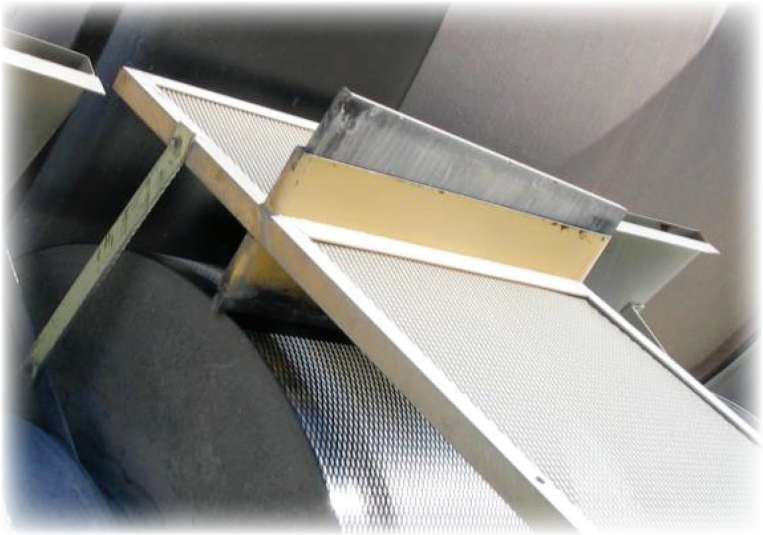
LE's Graphite Block is a unique carbon composition manufactured under the tightest internal controls. This makes for a highly consistent graphite block that guarantees optimum lubrication even in the most hostile environments, and prevents premature wear of the graphite.



LE's Graphite Blocks are cut to your specific requirements, ranging from 0.25-inch diameter up to 55-inch diameter rounds, and blocks as large as 16 inch thick by 32 inches wide and 110 inches long.

Beneficial Qualities

- **Custom** – Cut to your specific requirements to maximize benefits and performance.
- **Wear-Reducing** – Greatly reduces friction and wear.
- **Heat-Resistant** – Designed to perform in high-temperature applications.
- **Self-Extinguishing** – Will not sustain combustion under atmospheric conditions.
- **Rain-Resistant** – Does not wash off in rain.
- **Noise-Reducing** – Stops or substantially reduces high pitch squeal; and ensures smoother, quieter operation.
- **Clean** – Provides clean, dry lubrication.
- **Appearance** – Does not glaze over during operation.



On trunnion rollers, the block rides in a holder against the rotating surface, using gravity to allow the block's weight to deposit lubricant on the load-bearing surface. This maximizes the transference of the graphite against the load-bearing surface, minimizes roller and tire wear, and ensures optimal performance in all pressures and temperatures.

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Application Notes

- In kiln and dryer applications, the block lubricant transfers from the trunnion roller to kiln tire and lubricates both surfaces during normal rolling.
- In overhead and port crane applications, two blocks should be used per wheel. The wheels are double flanged, and a block must be on the leading and trailing wheels of each bogie.

	Part # Varies by Size & Shape
Size (custom)	Rounds: 0.25” to 55” diameter Rectangles: Up to 16” thick, 32” wide, 110” long
Appearance	Opaque
Color	Gray to black
Physical State	Solid (powder)
Odor	Odorless
Operating Temperature °C (°F)	-40 (-40) to 600 (1,112)
Melting Point °C (°F)	3,652-3697 (6,606-6,687)
Auto-Ignition Temperature, air method °C (°F)	300 (572)
Solubility	Insoluble
Surface Hardness, ASTM D2240	83-84
Coefficient of Friction	0.20
Wear Rate (Kilns)	Varies based on block thickness
Wear Rate (Cranes)	1” per 1,500 miles (25 mm per 2,400 km)
Quality Standards	Environmentally safe: non-GLP fish toxicity

The above are average values; minor variations that do not affect product performance are to be expected in normal manufacturing.

Typical Applications

- Trunnion rollers, riding rings and insert seals on rotary kilns, calciners and dryers where a dry lubricant is needed
- Free-floating support and thrust rollers and tires on rotary kilns and dryers
- Leading and training bogie wheels on overhead and port cranes
- Suitable for other industrial applications requiring dry film lubrication

Typical Industries

- Aggregate
- Asphalt
- Block Plants
- Cement
- Crushed Stone
- Fly Ash
- Grinding
- Lime
- Quarries
- Ready-Mix
- Sand & Gravel

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